Data-Efficient User Preference Acquisition for Shelf Rearrangement via Vision–Language Analysis and Interactive Ranking

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Personalized object arrangement in domestic environments requires understanding subjective user preferences that are often implicit and difficult to articulate. Existing approaches typically rely on user-generated demonstrations or curated datasets, which are costly to collect and do not scale well to real-world deployment. We present a data-efficient, human-centered pipeline for adapting shelf rearrangement models to individual users by learning preferences through natural interactions. The system acquires preference signals from user-provided reference photos and interactive ranking of shelf images, and uses a vision–language model to infer interpretable arrangement patterns such as spacing, grouping, and spatial bias. User preferences are represented as weighted, composable constraints that guide generative arrangement models. To improve adaptation efficiency over time, the system leverages a growing cross-user preference database to refine which examples are presented to new users and to warm-start preference inference. In addition, the pipeline supports adaptive discovery of new arrangement patterns when existing categories fail to explain user behavior. Together, these components enable transparent, scalable, and low-effort personalization of object arrangement goals suitable for deployment in household robotic and assistive systems.

1 Introduction

Household shelf rearrangement is a highly personalized task. Different users value accessibility, spacing, symmetry, density, and aesthetic style in different ways. For instance, a left-handed user may prefer objects packed toward the left side of a shelf, while another may prioritize evenly spaced or visually minimalist arrangements. For a robot deployed in a real home, adapting to such individual preferences is essential for usability and acceptance.

Existing approaches typically learn user preferences by training generative models on user-specific arrangement demonstrations, often using graph-based representations and variational autoencoders [5]. However, producing such demonstrations is burdensome for users, and many users struggle to articulate or construct their preferences explicitly. As a result, obtaining sufficient and representative preference data remains a key obstacle to deployment.

We argue that preference acquisition should leverage interactions users already find natural: looking at example shelves and expressing relative judgments. This paper explores two complementary mechanisms for collecting preference data: (1) analyzing user-provided photos of arrangements they already like, and (2) interactively querying users with ranked or paired shelf images. Both mechanisms produce sparse but informative signals that can be aggregated into structured preference representations.

2 Problem Statement

Let S denote a structured representation of a shelf environment, including geometry, compartments, and object categories. An arrangement X specifies object placements within S. Each user u is assumed to have an implicit utility function $\mathcal{U}_u(X)$ that evaluates how well an arrangement matches their preferences.

The objective is to generate an arrangement goal G such that

$$G^* = \arg\max_G \; \mathcal{U}_u(G \mid S), \tag{1}$$

subject to physical feasibility constraints.

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The challenge is that $\mathcal{U}_u$ is unknown and difficult to learn directly. We seek a method for acquiring preference data that is:

- **Low-effort**: requiring minimal cognitive and physical effort from users;
- **Expressive**: capturing both spatial and aesthetic preferences;
- **Interpretable**: producing preference representations that can be inspected and modified;
- **Deployable**: suitable for real-world household settings.

3 Related Work

3.1 Preference Learning from Human Feedback

Learning preferences from human feedback has been widely studied in machine learning and human–computer interaction. Early approaches rely on demonstrations or explicit labels, while more recent work emphasizes *comparative feedback*, such as pairwise rankings, which are often more reliable and less cognitively demanding than absolute scoring. Bradley–Terry and Thurstone-style models are commonly used to infer latent utilities from comparisons, and have been successfully applied in domains ranging from recommender systems to robotics. Recent work in reinforcement learning from human preferences further demonstrates that sparse comparisons can effectively guide complex models [2]. Our work builds on these ideas by using pairwise and lightweight scoring feedback to infer preferences over *arrangement patterns* rather than individual actions or trajectories.

3.2 Object Arrangement and Rearrangement in Robotics

Robotic object arrangement and rearrangement has primarily focused on physical feasibility, stability, and manipulation under clutter [3, 4]. Many systems assume that the target arrangement is specified a priori, or is derived from task semantics (e.g., setting a table). Some recent work explores learning spatial relations or scene graphs for arrangement generation, often using graph neural networks and generative models. However, these approaches typically require curated datasets or user-specific demonstrations and do not address how a deployed system should *acquire* personalized preferences efficiently. Our work complements this literature by focusing on preference acquisition and goal generation as a first-class problem.

3.3 Interactive Ranking and Active Querying

Interactive ranking systems are a common tool in recommender systems and preference elicitation. Active learning strategies select queries that maximize information gain or uncertainty reduction, thereby minimizing user effort. In HCI, such strategies are often combined with progressive disclosure and lightweight interfaces to improve usability. Our approach adopts active pair selection but extends it by leveraging a growing cross-user preference database, enabling population-informed query selection that accelerates adaptation for new users.

3.4 Population Priors and Cross-User Preference Modeling

Several lines of work show that aggregating preference data across users can improve cold-start performance. Collaborative filtering and Bayesian preference learning use population-level statistics as priors, which are refined through individual interactions. More recent systems cluster users by latent preferences and adapt query strategies accordingly. We follow a similar philosophy but operate at the level of *interpretable arrangement patterns*, allowing population statistics to inform both query selection and model initialization without obscuring individual differences.

3.5 Vision–Language Models for Visual Pattern Abstraction

Vision–language models (VLMs) have demonstrated strong capability in abstracting semantic and structural patterns from a small number of images [1, 6]. Unlike traditional visual classifiers, VLMs can express their interpretations in natural language, making them well suited for extracting human-interpretable arrangement patterns such as symmetry, spacing, or grouping. Recent work has explored VLMs for visual reasoning, style understanding, and few-shot concept discovery. Our work leverages these capabilities to analyze user-provided photos and retrieved shelf images, translating visual regularities into explicit preference constraints.

3.6 Preference-Aware Generative Models

Diffusion models and other modern generative techniques provide flexible mechanisms for encoding soft constraints and sampling diverse outputs [7]. Recent work explores composing multiple generative experts or energy terms to control generation. In parallel, constraint-based and modular generative approaches aim to improve interpretability and reuse. We build on these ideas by associating each arrangement pattern with a reusable diffusion expert, and selecting and weighting these experts based on inferred user preferences.

4 Preference Data Acquisition

We consider two complementary ways of collecting preference data when deploying a shelf rearrangement system to a specific user.

4.1 User-Provided Reference Photos

Users may provide photos of shelves or arrangements they personally find neat or appealing. These images may come from their own home, social media, or online sources.

Given a set of user-provided images $\mathcal{I}_u = \{I_1, \ldots, I_K\}$, we apply a vision–language model to extract shared arrangement patterns:

$$\widehat{\mathcal{P}}_u = \text{VLM\_}\textsc{Analyze}(\mathcal{I}_u). \tag{2}$$

The VLM outputs interpretable descriptors such as: *even spacing*, *left-biased packing*, *category grouping*, or *minimal clutter*. Because users select these images themselves, this signal is highly aligned with their latent preferences while requiring little effort.

4.2 Interactive Ranking and Comparison Interface

In cases where users cannot provide reference photos, or when additional clarification is needed, we employ an interactive ranking system.

The system presents users with:

- **Pairwise comparisons**: two shelf images are shown, and the user selects which arrangement they prefer.
- **Lightweight scoring**: users optionally assign “neatness” or “appeal” scores to individual images.

Each image is associated with known arrangement patterns. User responses generate preference constraints of the form:

$$I_a \succ I_b \implies \mathcal{P}(I_a) \succ \mathcal{P}(I_b), \tag{3}$$

where $\mathcal{P}(I)$ denotes the pattern set associated with image I.

4.3 Inferring Pattern Preferences from Feedback

We develop aggregation algorithms that map user choices to pattern-level preferences. Intuitively, if images exhibiting a particular pattern (e.g., even spacing) are consistently preferred, the weight of that pattern is increased.

4.4 Ranking Inference: From Pairwise Choices to Pattern Weights

Setup. Let $\mathcal{V} = \{c_1, \ldots, c_M\}$ be a fixed vocabulary of arrangement patterns (constraints). Each candidate shelf image i is annotated (or automatically predicted) with a binary pattern indicator vector $x_i \in \{0,1\}^M$, where $(x_i)_m = 1$ indicates that pattern c_m is present in image i. We learn a user-specific weight vector $w \in \mathbb{R}^M$ such that higher w_m implies stronger preference for pattern c_m. We define the latent utility of image i as

$$u_i \;=\; w^\top x_i. \tag{4}$$

We observe pairwise comparisons of the form $(a \succ b)$ meaning the user prefers image a over image b. Let $\mathcal{D}_{\text{pair}} = \{(a_k, b_k)\}_{k=1}^K$ denote the set of comparisons.

Bradley–Terry (logistic) likelihood. We model the probability that the user prefers a over b using a Bradley–Terry style logistic choice model:

$$\Pr(a \succ b \mid w) \;=\; \sigma(u_a - u_b) \;=\; \frac{1}{1 + \exp\big(-(w^\top(x_a - x_b))\big)}, \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function.

The negative log-likelihood over comparisons is

$$\mathcal{L}_{\text{pair}}(w) \;=\; -\sum_{k=1}^{K} \log \sigma\big(w^\top(x_{a_k} - x_{b_k})\big). \tag{6}$$

Optional rating supervision. If the interface also collects a coarse rating $r_i \in \{1, \ldots, 5\}$ for some images, we can convert ratings to (i) pairwise constraints by comparing rated images, or (ii) a regression term. For the regression option, map ratings to a centered real value $\tilde{r}_i$ (e.g., $\tilde{r}_i = r_i - 3$) and minimize:

$$\mathcal{L}_{\text{rate}}(w) \;=\; \sum_{i \in \mathcal{D}_{\text{rate}}} \big(w^\top x_i - \beta \tilde{r}_i\big)^2, \tag{7}$$

where β is a scale factor (learned or fixed).

Regularization / prior (MAP estimation). To stabilize learning with few interactions, we use an ℓ_2 prior:

$$p(w) = \mathcal{N}(0, \lambda^{-1} I) \quad \Longleftrightarrow \quad \mathcal{L}_{\text{prior}}(w) = \frac{\lambda}{2}\|w\|_2^2. \tag{8}$$

The overall objective is

$$\min_{w} \; \mathcal{L}(w) = \mathcal{L}_{\text{pair}}(w) + \gamma \mathcal{L}_{\text{rate}}(w) + \mathcal{L}_{\text{prior}}(w), \tag{9}$$

where $\gamma \geq 0$ balances ratings (if present).

Gradient and online updates (SGD).. For a single comparison $(a \succ b)$, let $\Delta x = x_a - x_b$ and $z = w^\top \Delta x$. The gradient of the pairwise loss term is:

$$\nabla_w\big(-\log \sigma(z)\big) \;=\; -(1 - \sigma(z))\,\Delta x. \tag{10}$$

Including ℓ_2 regularization, an online SGD update with step size η is:

$$w \leftarrow w - \eta\Big(- (1 - \sigma(z))\Delta x + \lambda w\Big). \tag{11}$$

This allows preference weights to be updated interactively after each user choice.

From weights to selected constraints. After learning w, we produce a sparse constraint set by selecting the top-m patterns or those above a threshold:

$$\widehat{C}_u \;=\; \{c_m \in \mathcal{V} \;:\; w_m \geq \tau\} \quad \text{or} \quad \textsc{TopK}(w). \tag{12}$$

The resulting $\widehat{C}_u$ and corresponding normalized weights $\tilde{w}_m$ are then used to select and weight the constraint-specific diffusion experts in composition.

4.4.1 Active Pair Selection to Minimize User Effort. To reduce the number of comparisons needed, we choose which pair of images to ask next. Intuitively, comparisons are most informative when the model is uncertain. For a candidate pair (i, j), the model predicts

$$p_{ij} = \Pr(i \succ j \mid w) = \sigma\big(w^\top(x_i - x_j)\big). \tag{13}$$

We select the next query by maximizing uncertainty (entropy):

$$(i^*, j^*) = \arg\max_{(i,j)\in Q} H(p_{ij}) = \arg\max_{(i,j)\in Q} \Big[- p_{ij}\log p_{ij} - (1 - p_{ij})\log(1 - p_{ij})\Big], \tag{14}$$

where Q is a pool of candidate pairs.

A practical alternative is to choose pairs with utilities close together:

$$(i^*, j^*) = \arg\min_{(i,j)\in Q} |u_i - u_j|. \tag{15}$$

Either strategy focuses user interactions on comparisons that refine pattern weights fastest.

4.4.2 Algorithm: Preference Weight Learning from Interactive Ranking.

Input: Pattern vocabulary $\mathcal{V}$; image pool $\{x_i\}_{i=1}^N$; optional ratings $\mathcal{D}_{\text{rate}}$; regularization λ; stepsize η; interaction budget K.
Output: User pattern weights w; selected constraints $\widehat{C}_u$.

(1) Initialize $w \leftarrow 0$ (or prior mean).
(2) (Optional) If ratings exist, perform a few gradient steps on $\mathcal{L}_{\text{rate}}(w) + \frac{\lambda}{2}\|w\|^2$.
(3) For $k = 1$ to K:
  (a) Select query pair (i, j) using entropy (14) (or margin (15)).
  (b) Ask user to choose preferred image; observe outcome $(a \succ b)$.
  (c) Compute $\Delta x = x_a - x_b$, $z = w^\top \Delta x$.
  (d) Update weights using SGD (11).
(4) Produce $\widehat{C}_u$ by thresholding or Top-m selection (12).

Interpretation. The learned w directly indicates which arrangement patterns are preferred. These weights can (i) drive explainable UI summaries (“you prefer even spacing and low clutter”), (ii) select constraint-specific diffusion experts, and (iii) weight their composition during sampling.

Let $\mathcal{V}$ be a vocabulary of patterns. We estimate user-specific pattern weights:

$$w_c \propto \sum_{(I_a \succ I_b)} \mathbb{I}[c \in \mathcal{P}(I_a)] - \mathbb{I}[c \in \mathcal{P}(I_b)], \tag{16}$$

optionally combined with absolute scores. This yields a ranked or weighted pattern set $\widehat{C}_u$.

4.5 Population-Informed Querying with a Preference Database

Motivation. As the system is deployed, we accumulate a database of user interactions—pairwise choices, per-image scores, and the inferred pattern weights. We use this database to (i) improve which images are shown to new users, (ii) reduce the number of queries needed to identify preferences, and (iii) increase the diversity and diagnostic power of the candidate image pool.

Preference database. Let $\mathcal{B}$ denote a database of U historical users. For each user u, we store:

$$\mathcal{B} = \left\{ \left(\mathcal{D}^{(u)}_{\text{pair}}, \mathcal{D}^{(u)}_{\text{rate}}, \hat{w}^{(u)} \right) \right\}_{u=1}^{U}, \tag{17}$$

where $\mathcal{D}^{(u)}_{\text{pair}}$ are pairwise comparisons, $\mathcal{D}^{(u)}_{\text{rate}}$ are optional ratings, and $\hat{w}^{(u)}$ is the learned pattern weight vector from Section 4.4. Each image i in the library has a pattern indicator $x_i \in \{0,1\}^M$ (manual labels or predicted by a VLM).

Learning a population prior for warm start. A simple way to leverage $\mathcal{B}$ is to construct a population prior over w:

$$\mu = \frac{1}{U} \sum_{u=1}^{U} \hat{w}^{(u)}, \qquad \Sigma = \mathrm{Cov}\left(\{\hat{w}^{(u)}\}_{u=1}^{U} \right), \tag{18}$$

and initialize a new user's weights with $w_0 \leftarrow \mu$ (or a sample from $\mathcal{N}(\mu, \Sigma)$). This warm start reduces early uncertainty and can lower the number of required interactions.

Clustering users to personalize the query pool. Because preferences are multi-modal across the population, we optionally cluster users in weight-space:

$$\{\hat{w}^{(u)}\}_{u=1}^{U} \xrightarrow{\textsc{Cluster}} \{(\mu_g, \Sigma_g)\}_{g=1}^{G}, \tag{19}$$

and assign a new user to a cluster using a small number of initial queries (Section 4.4.1). We then use the corresponding cluster prior $\mathcal{N}(\mu_g, \Sigma_g)$ to focus subsequent queries.

Selecting informative images for new users. Rather than sampling images uniformly, we curate a candidate set that is both *diverse in patterns* and *diagnostic*. Let π be a distribution over patterns estimated from $\mathcal{B}$ (e.g., frequency or entropy-adjusted frequency). We define an image selection score that favors images covering under-represented or high-uncertainty patterns:

$$\textsc{CoverScore}(i) = \sum_{m=1}^{M} (x_i)_m \cdot \rho_m, \quad \rho_m \propto \underbrace{H_m}_{\text{informativeness}} \cdot \underbrace{(1/\mathrm{freq}_m)}_{\text{coverage}}, \tag{20}$$

where freq_m is how often pattern m appears in the database's presented images, and H_m can be set to the empirical uncertainty of pattern m (e.g., variance of $\hat{w}^{(u)}_m$ across users). We construct a query pool Q by selecting top images under (20) while enforcing diversity constraints (e.g., do not select near-duplicates in image embedding space).

Population-informed pair selection. Given a selected image pool, we refine which *pairs* to ask by combining (i) per-user uncertainty (entropy) with (ii) expected discriminability across the population. For a candidate pair (i, j) define:

$$p_{ij}(w) = \sigma\big(w^\top (x_i - x_j)\big), \tag{21}$$

$$H_{ij}(w) = -p_{ij}(w) \log p_{ij}(w) - (1 - p_{ij}(w)) \log(1 - p_{ij}(w)), \tag{22}$$

$$\textsc{PopDisc}(i, j) = \mathrm{Var}_{u\sim\mathcal{B}}\left[\sigma\big((\hat{w}^{(u)})^\top (x_i - x_j)\big)\right]. \tag{23}$$

We then score pairs for a new user by:

$$\textsc{PairScore}(i, j) = \underbrace{H_{ij}(w)}_{\text{current-user uncertainty}} + \kappa \underbrace{\textsc{PopDisc}(i, j)}_{\text{population discriminability}}, \tag{24}$$

and query the pair with the largest $\textsc{PairScore}$:

$$(i^*, j^*) = \arg\max_{(i,j)\in Q} \textsc{PairScore}(i, j). \tag{25}$$

Intuitively, $\textsc{PopDisc}$ prefers pairs that different user types disagree on, making them especially helpful for quickly identifying which preference region a new user belongs to.

Using scores to calibrate pattern strength. If the interface collects absolute "neatness" or "appeal" scores, we can refine image selection by learning global score calibration per pattern. Let $\bar{r}_i$ be the mean rating of image i across users. A simple calibration model is:

$$\bar{r}_i \approx a + b^\top x_i, \tag{26}$$

where b_m indicates how strongly pattern m tends to increase perceived neatness in the population. For new users, b can serve as a prior over likely-positive patterns, while interactive ranking identifies the user-specific deviations.

Algorithm: Database-Refined Image Curriculum for New Users.

Input: Preference DB $\mathcal{B}$; image library $\{x_i\}_{i=1}^N$; interaction budget K; weights κ; diversity constraint.
Output: Adaptive queries (pairs) for the new user; updated user weights w.

(1) Build population statistics (μ, Σ) or clusters $\{(\mu_g, \Sigma_g)\}$ from $\{\hat{w}^{(u)}\}$.
(2) Select a diverse image pool using $\textsc{CoverScore}$ (20) (and deduplicate via embeddings).
(3) Initialize $w \leftarrow \mu$ (or the most probable cluster mean).
(4) For $k = 1$ to K:
  (a) Choose next pair (i, j) by maximizing $\textsc{PairScore}$ (24).
  (b) Query the user; update w with the online SGD step from Eq. (9) in Section 4.4.
  (c) (Optional) Re-assign cluster if using clustered priors.
(5) Return the refined constraint set $\widehat{C}_u$ and weights w for downstream goal generation.

Practical notes. This population-informed strategy does not require sharing raw user images: the database can store only interaction outcomes and pattern vectors. It also naturally improves over time: as $\mathcal{B}$ grows, the system learns which queries are most diagnostic and reduces required user effort.

4.6 Adaptive Discovery of New Arrangement Patterns

While the system initially relies on a predefined vocabulary of arrangement patterns (e.g., even spacing, symmetry, left-biased packing), real users may exhibit preferences that are not well captured by this fixed set. To address this limitation, we extend the proposed pipeline with an adaptive mechanism for discovering new patterns over time.

Motivation. Household arrangements exhibit long-tail variability driven by personal habits, cultural norms, and idiosyncratic aesthetics. Forcing all preferences into a closed pattern vocabulary risks mischaracterizing user intent and limits expressiveness. Instead, we treat the pattern vocabulary as *open and extensible*, allowing the system to propose new candidate patterns when existing ones are insufficient.

Detecting unexplained preference signals. During interactive ranking, the system monitors residuals between predicted and observed user choices. Let $p_{ab} = \sigma(w^\top (x_a - x_b))$ be the predicted preference probability for a queried pair (a, b). Consistently high prediction error,

$$\mathbb{E}[|\mathbb{I}(a \succ b) - p_{ab}|] \;>\; \epsilon, \tag{27}$$

indicates that the current pattern set $\mathcal{V}$ fails to explain the user's preferences. Such mismatches trigger a pattern discovery process.

Candidate pattern proposal via VLM abstraction. When unexplained preferences are detected, we collect a small set of images that the user consistently prefers but that are poorly explained by existing patterns. A vision–language model is then prompted to describe recurring visual or structural regularities across these images:

$$c_{\text{new}} = \text{VLM\_}\textsc{Propose}\big(\{I_i : \text{high residual}\}\big). \tag{28}$$

The output is a natural-language pattern description (e.g., “objects aligned flush to the front edge” or “dense clustering with visual anchors”).

Pattern validation and grounding. Proposed patterns are not immediately added to the vocabulary. Instead, they undergo a lightweight validation step:

- **User confirmation:** the system presents a short explanation and a few representative examples and asks whether the pattern reflects the user's intent.
- **Population support:** the system checks whether similar residual patterns appear across multiple users in the preference database.

Only patterns that pass one or both checks are admitted into the extended vocabulary $\mathcal{V} \leftarrow \mathcal{V} \cup \{c_{\text{new}}\}$.

Bootstrapping generative support. Once a new pattern is accepted, the system bootstraps generative support in one of two ways:

(1) **Soft constraint approximation:** the pattern is initially implemented as a rule-based or energy-based constraint derived from the VLM description.
(2) **Deferred expert training:** if sufficient supporting examples accumulate, a new constraint-specific diffusion expert $\mathcal{D}_{c_{\text{new}}}$ is trained offline and added to the expert library.

Until a dedicated expert is available, the new pattern can be composed with existing experts as a weak guiding signal.

Continual evolution of the pattern space. Over time, the pattern vocabulary evolves as the system is deployed to more users. Frequently confirmed patterns become first-class constraints with dedicated generative experts, while rarely used or redundant patterns may be merged or deprecated. This continual refinement allows the system to balance structure and flexibility, maintaining interpretability while expanding expressive power.

Integration with ranking inference. Newly discovered patterns are incorporated into the ranking inference model by augmenting the pattern indicator vectors x_i. Subsequent comparisons naturally update their associated weights, allowing the system to quantify how strongly each user prefers the new pattern relative to existing ones.

5 From Preferences to Arrangement Goals

5.1 Preferences as Composable Constraints

User preferences are represented as a sparse set of constraints:

$$\widehat{C}_u \subset \mathcal{V}, \tag{29}$$

where each constraint corresponds to a reusable arrangement pattern.

5.2 Constraint-Specific Diffusion Models

For each pattern $c \in \mathcal{V}$, we assume a pre-trained diffusion model $\mathcal{D}_c$ that encodes how that pattern shapes valid arrangements [7].

5.3 Composing Constraints for Goal Generation

Selected constraints are composed by combining their score functions:

$$s_{\text{comp}}(X, t \mid S) = \sum_{c \in \widehat{C}_u} w_c \, s_{\theta_c}(X, t \mid S), \tag{30}$$

and sampled to produce candidate arrangement goals.

5.4 Final Goal Selection

Candidate goals are filtered for feasibility and preference alignment:

$$G^* = \arg\max_{G_j} \left[\alpha \, \textsc{Feasible}(G_j \mid S) + (1 - \alpha) \, \textsc{PrefScore}(G_j \mid \widehat{C}_u) \right]. \tag{31}$$

6 Conclusion

We presented a practical framework for acquiring user preference data for shelf rearrangement through user-provided photos and interactive ranking. By translating sparse human feedback into pattern-level constraints, the system enables efficient adaptation of generative arrangement models to individual users with minimal effort.

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